Journal of Electrical Engineering & Technology (2026) 21:55–69
https://doi.org/10.1007/s42835-025-02394-0

ORIGINAL ARTICLE

Robust Optimal Scheduling of Microgrid Based on Data-Driven Segmented Power Multi-Interval Uncertainty Sets

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Received: 11 November 2024 / Revised: 16 July 2025 / Accepted: 4 August 2025 / Published online: 18 August 2025
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Abstract
The uncertainties of renewable distributed generation power output and load demand have created significant challenges to the scheduling of grid-connected microgrids and have restricted the carbon emission reduction capability of microgrid. To address the problems posed by source-load uncertainties of microgrid and carbon emissions on scheduling, this paper proposes a two-stage robust optimal scheduling model for microgrid based on data-driven segmented multi-interval uncertainty sets. Firstly, the source-load predicted power and error data are segmentally fitted based on a Gaussian mixture model, which is combined with a box-type uncertainty set to establish a multi-interval uncertainty set for the power partition segments, which is used to characterize the wind, photovoltaic and load uncertainties more accurately; then, a robust optimal scheduling model for microgrid considering segmented multi-interval uncertainty sets and carbon emissions is constructed to minimize the microgrid operation cost and carbon emission cost. The model is solved using an improved Columns and Constraints Generation (CC&G) algorithm. Finally, the effectiveness of the proposed model is verified by case studies. The results show that the total cost of the proposed method is reduced by 8.15%, 3.13%, and 12.10% compared to single-interval RO, multi-interval RO, and ellipsoidal RO, respectively, and the solution time is reduced by about 100 s compared to SO. This indicates that the proposed model can better balance the economy and robustness of MG. In addition, the robustness and economy of the scheduling scheme can be flexibly adjusted by choosing appropriate uncertain budget parameters as needed during the actual formulation of the scheduling scheme.

Keywords Microgrid · Data-driven · Segmented multi-interval · Robust optimization · Uncertainty

1 Introduction

Microgrid (MG), which are small-scale power generation and distribution networks composed of renewable energy (RE), energy storage systems, and loads, plays an important role in the energy structure transition of the power system [1, 2]. MG creates significant economic benefits for the power system, such as reducing the grid energy losses and improving the system efficiency. However, the uncertainty of RE power outputs and load demand brings great challenges to the reliable operation of MG [3, 4]. How to cope with the uncertainty that exists in MG and realize the stable and economic operation of MG has been a worthwhile research problem [5].

1.1 Literature Review

In recent years, most scholars have conducted deep studies on the optimization of MG scheduling taking into account

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the influence of uncertainty factors with the objectives of operation economy, RE absorption capacity, and reduction of carbon emissions [6–8]. Currently, there are scenario analysis, stochastic optimization (SO), robust optimization (RO), and distributed robust optimization (DRO) approaches for uncertainty factors [9–11]. Among them, both the scenario analysis method and the SO method depend on information about the probability distribution of uncertainty variables, which requires a quantity of historical sample data and increases the complexity of the problem, while some extreme scenarios may be ignored. [12] proposes a stochastic model predictive control framework that accounts for source-side uncertainty, which preserves both upward and downward adjustment capabilities of the source. [13] introduces a stochastic multi-objective optimization model for grid-connected imbalance MGs, utilizing a roulette mechanism to generate uncertainty scenarios for the source, load, and electricity tariffs. RO does not depend on the probability distribution of uncertainties, which expresses the uncertainty of variables in the form of interval sets, and is capable of solving the optimal scheduling scheme of the system under extreme scenarios, and thus has been widely used in MG optimal scheduling [14, 15.16] constructed a two-stage robust energy management model for MG with polyhedral uncertainty sets to describe the uncertainties of source and load. [17] formulated a two-stage RO model with binary recourse variables based on the polyhedral uncertainty set of periodic budgets. However, all of the above literature used simple uncertainty sets to describe the uncertainty of MG, which leads to many scenarios in the set that are impossible in practice, making the dispatch results too conservative and reducing the operational efficiency of the power system [18, 19] .

To overcome the specificity of SO and the conservative defects of RO, DRO extracts information about probability from historical data and constructs an ambiguous set containing probability distributions of uncertain variable factors, which in turn solves the worst case minimum expected cost [20]. According to the difference of an ambiguous set construction approach, DRO can be categorized into two types: moment information-based and distance metric-based. The DRO method based on moment information derives probability distribution parameters for uncertain parameters from historical data [21]. The distance-based DRO method builds ambiguity sets by measuring the statistical distance between the true and hypothetical distribution [22]. However, to improve data processability, the methods described above convert the DRO model to a convex problem through approximation or relaxation, which may lead to suboptimal results and thus affect the robustness of the scheduling results. So, the DRO technique may not be the best approach to cope with the uncertainty of MG operators' concerns.

Although the above-mentioned literature has achieved fruitful outcomes in terms of reducing RO conservativeness, most of the studies have ignored the direct relationship between uncertainty parameter datasets and prediction errors. It remains a challenge to deeply explore the features embedded in the historical data and integrate them into the construction of the uncertainty set thereby enhancing the quality of the optimization results. [23] constructed uncertainty sets containing the spatiotemporal correlation of PV power generation by combining least squares fitting straight lines with polyhedral uncertainty sets. [24, 25] utilized multi-interval uncertainty sets to solve the adaptive RO scheduling problem in MGs, which to some extent avoids the worst-case scenarios always located at the boundaries of the prediction intervals. [26, 27] fitted uncertainty sets by using a mixture model of the Dirichlet process and a Bayesian nonparametric approach, which eliminated the discrepancy between traditional polyhedral uncertainty sets and historical scenarios. [28, 29] described the spatio-temporal correlation of uncertainty sets by combining certain extreme scenarios in selected historical data with traditional convex uncertainty sets and proved their applicability. [30] combines temporal correlation with interval partitioning to characterize the uncertainty of source loads, which reduces the conservativeness of the optimal solution for MG scheduling.

1.2 Motivation and Contribution

In summary, this study aims to address the optimal scheduling of microgrids under the modeling of multiple uncertainties and carbon emission issues, to minimize both operational and environmental costs. Given the limitations of existing optimization models in handling uncertainty sets, a robust optimal scheduling model based on data-driven segmentation of multi-interval uncertainty sets is proposed to solve the MG scheduling problem. Table 1 provides a comprehensive comparison of this work with previous research in the field. Two key insights can be drawn from these comparisons. First, the direct relationship between uncertainty parameter datasets and prediction errors has not been adequately addressed. Second, the issue of carbon emissions has been overlooked by some studies, such as those in [14, 16].

The main contributions of this work are summarized as follows:

(1) A new approach for modeling multi-interval uncertainty sets in power-partitioned segments is proposed. In aiming to reduce the conservativeness of robust optimal scheduling strategies and improve the economics of

Table 1 Comprehensive comparison of this work with previous research

Ref	Carbon emissions	Demand response	Wind	PV	Load	Uncertainty modeling methods	Solution Methods
[9]	✓	×	✓	✓	×	SO	CVX
[12]	×	✓	✓	✓	×	SO	TOMLAB
[13]	×	×	✓	✓	✓	SO	Knitro
[16]	×	×	✓	✓	×	RO	C&CG+GUROBI
[17]	×	×	✓	×	×	RO	C&CG+CPLEX
[18]	×	×	✓	✓	✓	ARO	ATC+C&CG
[21]	×	×	✓	✓	×	DRO	C&CG
[22]	×	×	✓	✓	×	DRO	GUROBI
[23]	×	×	✓	✓	×	ARO	C&CG
[24]	×	×	✓	✓	✓	RO	C&CG
[26]	✓	×	✓	✓	✓	RO	C&CG
[28]	×	×	×	✓	✓	RO	C&CG
[29]	×	×	✓	✓	×	Data-driven RO	CPLEX
[30]	×	✓	✓	✓	✓	Data-driven RO	C&CG
This work	✓	✓	✓	✓	✓	Data-driven RO	C&CG

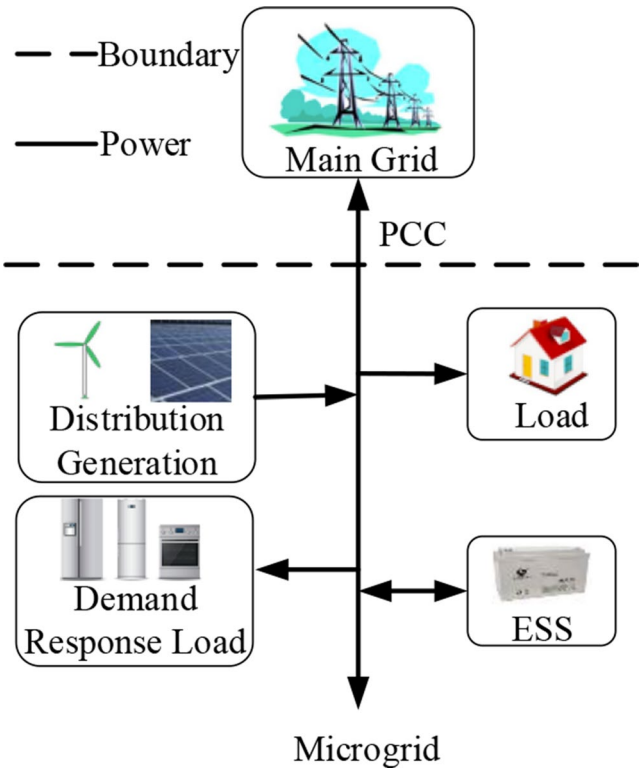


Fig. 1 Microgrid system structure

MG scheduling, a Gaussian mixture model is used to fit the prediction errors within different power segments based on the historical predicted power and error data of the uncertain parameters. Then, a new power segmented multi-interval uncertainty set modeling method is constructed based on the error distribution fitting results.

(2) A two-stage RO scheduling model for microgrids that combines source-load uncertainty and environmental costs is established. The model considers the operating economy of the microgrid, environmental protection, and the conservativeness of the optimization results.

(3) The proposed model is linearized by the Big M method and the Lagrangian dual theory, and the improved CCG algorithm is used to solve the MILP. four cases are designed for comparative analysis, which verifies that the proposed model balances the economic and environmental benefits of MG operation; while guaranteeing the robustness of the scheduling strategy and the consumption of renewable energy.

The rest of the paper is organized as follows: the second section describes the data-driven segmented multi-interval uncertainty set (DDSMIUS) based modeling process proposed in this paper. Section 3 develops a two-stage RO model based on DDSMIUS. The solution procedure of the two-stage RO model is given in Sect. 4. Case study and simulation results are given and discussed in Sect. 5, followed by conclusions in Sect. 6.

2 MG System Structure

Figure 1 demonstrates MG system structure.. The MG comprises controllable distributed power sources (microturbines), uncontrollable renewable generation sources (PV, WT), energy storage systems (ESS), and loads. The MG interacts with the upper distribution grid regarding energy through a centralized power tie line. The control center optimizes the scheduling of each flexible unit in the system to improve the economy and reliability of operation while considering the economy and environmental friendliness of the MG.

3 Data-Driven Segmented Multi-Interval Uncertainty Set-Based Modeling

Currently, the MG robust scheduling ignores the important parameter characteristics of the source and loads historical data, thus splitting the connection between the day-ahead prediction and scheduling. Therefore, to improve the shortcomings of RO which is too conservative, firstly, the prediction errors in different power intervals are fitted by utilizing the information of probabilistic characteristics possessed by the wind and photovoltaic power outputs and the loads themselves, and it is confirmed that the method of power segment fitting can reflect the real distribution of prediction errors more accurately. Then, the traditional single interval uncertainty set is divided into multiple intervals based on the fitting results of the error distribution corresponding to different power segments. Finally, the uncertainty budget parameters are assigned according to the distribution of prediction errors in each power segment, making the defined multi-interval uncertainty set more consistent with the actual situation and effectively reducing the scheduling plan's conservativeness.

3.1 Relationship Between the Predicted Power of the Source-Load and Its Error

The actual values of wind, photovoltaic, and load power are represented by the sum of their power prediction values and the prediction error:

$$e_{\psi,t}=P_{\psi,t}-\bar{P}_{\psi,t} \tag{1}$$

where cell ψ includes wind turbine(WT), photovoltaic(PV), and load; $e_{\psi,t}$ is the power prediction error of the cell ψ at time t; $P_{\psi,t}$ is the actual power of the cell ψ at time t; $\bar{P}_{\psi,t}$ is the power prediction value of the cell ψ at time t.

[31] pointed out that there are differences in the probability distributions of forecast errors corresponding to power forecast values located in different segments. Taking wind power as an example, the historical measured data of Elia grid from January 1, 2022, to December 31, 2022, is selected as the object of analysis. The data were normalized by the installed wind power capacity. The scatter plot of the wind power prediction value and its prediction error is shown in Fig. 2, from which it can be seen that the distribution of prediction error corresponding to the low power prediction value is more centralized and smaller in value due to the influence of uncertainty factors such as meteorological prediction error and micro-terrain difference, while the prediction error corresponding to the high power prediction value is more scattered and larger in value. This indicates that the distribution characteristics of the power prediction errors are correlated with the magnitude of the power prediction values, and these characteristics need to be mined to better characterize the source-load uncertainties.

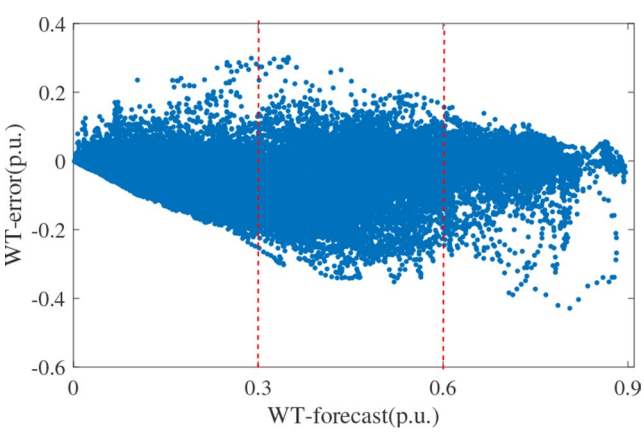


Fig. 2 Scatter plot of wind power prediction and its prediction error

Gaussian mixture model has the advantage of describing different characteristic distributions with high applicability, and can effectively deal with the strong uncertainty of RE and load forecast error probability distribution fitting problem, as shown in Eq. 2.

$$\begin{aligned} &f^i(\Delta P)=\sum_{k=1}^{m}\omega_k f_{\mathrm{N}}(\Delta P \mid \mu_k,\sigma_k^2) \\ &0<\omega_k<1,\ \omega_1+\omega_2+\ldots+\omega_n=1 \end{aligned} \tag{2}$$

where $f^i(\Delta P)$ denotes the probability density function of the power prediction error ΔP in the *ith* power segment; ω_k is the weight coefficient of the *kth* normal distribution; $f_{\mathrm{N}}(\Delta P \mid \mu_k,\sigma_k^2)$ is the probability density function of the *kth* normal distribution; μ_k,σ_k^2 is the expectation and variance, respectively; m is the number of normal distributions in the Gaussian mixture model.

In this paper, the source-load power prediction values are divided into three segments, specifically [0,0.3], [0.3,0.6], and [0.6,1], and the Gaussian mixture model is used to fit the power prediction errors within each segment, respectively. The probability distribution of the prediction error within each interval and the probability distribution of the overall error without segmentation are shown in Fig. 3. As can be seen in Fig. 3, the probability distribution of prediction error within different segments has significant differences: when the predicted value is less than 0.3, the fitted curve of the probability distribution of prediction error has obvious spiky characteristics; when the WT power is between 0.3 and 0.6, the fitted curve shows symmetry; and when the WT power is greater than 0.6, the fitted curve has obvious right skewed characteristics. Therefore, the method of fitting the prediction error by segments of the predicted power can reflect the distribution of the prediction error more realistically.

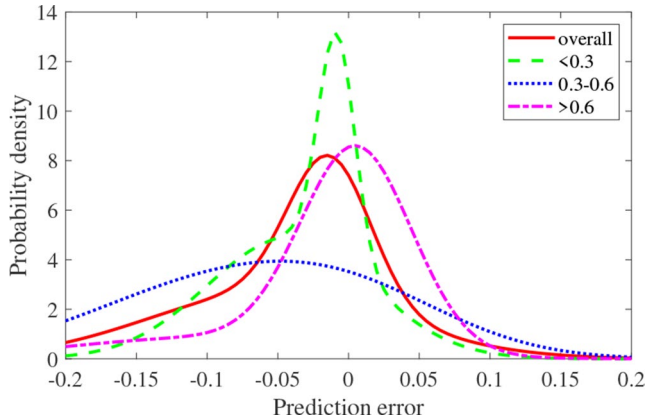


Fig. 3 Probability density fitting curve for wind power prediction error

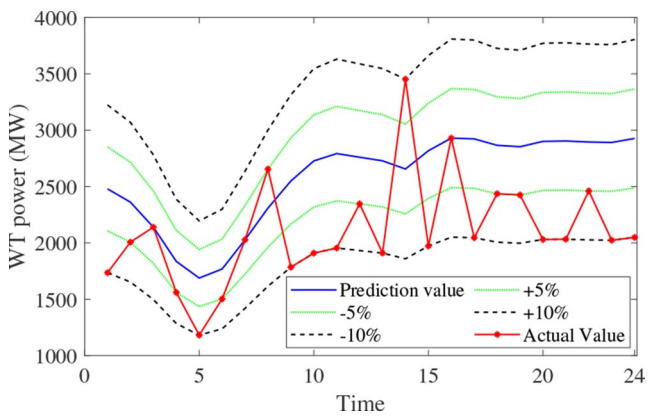


Fig. 4 Multi-interval uncertainty set

3.2 Multi-Interval Uncertainty Set Modeling for Power Segmentation

Robust optimal scheduling models often use a single interval uncertainty set to describe the source load uncertainty, but the optimization results are more conservative. To this end, a multi-interval uncertainty set is introduced in [24], as shown in Fig. 4. As can be seen from Fig. 4, although the multi-interval source-load uncertainty set can reduce the conservativeness of robust optimization to a certain extent, it ignores the relationship between the predicted power of the source-load and the prediction error, which leads to the uncertainty set containing a multitude of extreme scenarios that are unlikely to occur in practice.

To effectively eliminate the unlikely extreme scenarios on the uncertainty sets, a more realistic multi-interval uncertainty set for power segmentation is constructed by considering the relationship between the historical predicted power and the prediction error.

First, the probability distribution of the corresponding prediction error is obtained based on the source-load prediction power segmentation condition;

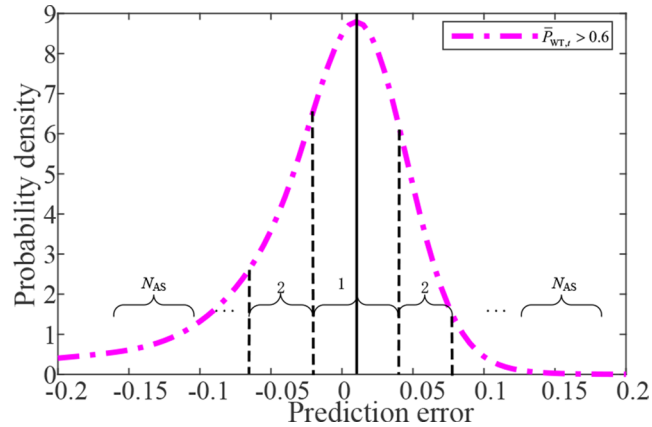


Fig. 5 Multi-interval division of power segmentation uncertainty set

Then, taking the probability distribution of the prediction error with the power prediction scalar value located in the interval [0.6,1] as an example, the construction method of the multi-interval uncertainties set of the power segmentation error is further analyzed in conjunction with Fig. 5;

Finally, based on the magnitude of prediction error and the probability of occurrence, combined with the uncertainty budget parameter Γ_ψ the single interval of error $[\bar{P}_{\psi,t,s}-e^-_{\psi,t,s}\ \bar{P}_{\psi,t,s}+e^+_{\psi,t,s}]$ is categorized into N_{AS} intervals, which satisfy Eq. 3.

$$\begin{cases} \sum_{s=1}^{N_{\mathrm{SP}}}\sum_{j=1}^{N_{\mathrm{AS}}}\left(\frac{e^{j+}_{\psi,t,s}+e^{j-}_{\psi,t,s}}{e^{+}_{\psi,t,s}+e^{-}_{\psi,t,s}}\Gamma_{\psi,j,s}\right)=\Gamma_\psi \\ \frac{\rho_{\psi,j}}{\rho_\psi}=\frac{\Gamma_{\psi,j}}{\Gamma_\psi} \end{cases} \tag{3}$$

where N_{SP} is the number of power prediction segments; $e^{j+}_{\psi,t,s}$ and $e^{j-}_{\psi,t,s}$ are the upper and lower deviation values of the prediction power in the s segment j interval of the cell ψ at t; $\Gamma_{\psi,j,s}$ is the uncertainty time budget parameter in the s segment j interval of the cell ψ at t, which can be adjusted to change the degree of uncertainty; $e^{+}_{\psi,t,s}$ and $e^{-}_{\psi,t,s}$ are the upper and lower deviation values of the prediction of the cell ψ at t in a single interval; Γ_ψ is the uncertainty time budget parameter; $\rho_{\psi,j}$ is the probability of occurrence in the j interval of the cell ψ, the value of which can be obtained by fitting the corresponding probability distribution. And ρ_ψ is the probability of occurrence of the cell ψ on a single interval.

Based on the constructed data-driven segmented multi-interval source load uncertainty set, the RE and load uncertainties in MG are shown in Eq. 4, and this uncertainty scenario is shown in Fig. 6.

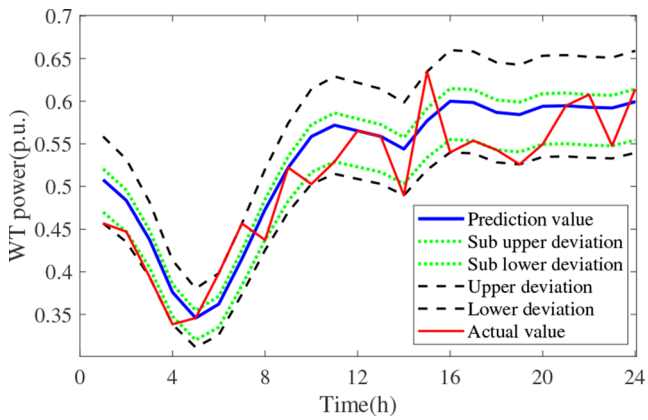


Fig. 6 Segmented multi-interval uncertainty set

$$U_\psi = \begin{cases} P_{\psi,t,s} = \bar{P}_{\psi,t,s} + \\ \sum_{s=1}^{N_{SP}} \sum_{j=1}^{N_{AS}} \left(e^{+}_{\psi,t} \varepsilon^{j+}_{\psi,t} - e^{-}_{\psi,t} \varepsilon^{j-}_{\psi,t} \right) \\ 0 \le \varepsilon^{j\pm}_{\psi,t,s} \le 1 \\ 0 \le \sum_{j=1}^{N_{AS}} \left(\varepsilon^{j+}_{\psi,t,s} - \varepsilon^{j-}_{\psi,t,s} \right) \le 1 \\ \sum_{t=1}^{T} \left(\varepsilon^{j+}_{\psi,t,s} - \varepsilon^{j-}_{\psi,t,s} \right) \le \Gamma_{\psi,j} \end{cases} \tag{4}$$

where: U_ψ represents the segmented multi-interval uncertainties set of the cell ψ; $\varepsilon^{j+}_{\psi,t,s}$ and $\varepsilon^{j-}_{\psi,t,s}$ are Boolean variables, indicating that the prediction bias variable is located in the upper and lower half of the *jth* interval.

In comparison to the uncertainty set introduced in [24], the prediction errors corresponding to the power values in different segments of the uncertainties set proposed are determined according to the parameters of the fitted probability distribution, which makes it closer to reality, and thus the conservativeness of the RO results can be reduced. This is attributed to the fact that this paper not only considers the time dependence of the source and loads but also the distribution characteristics of the prediction errors corresponding to their day-ahead prediction values.

4 Mathematical Modeling of MG Systems

4.1 Controllable Distributed Generation

The cost of controllable distributed generation in MG is shown in Eq. 5, which includes operation and maintenance costs and fuel consumption costs.

$$C_{MT}(t) = b_{MT} P_{MT}(t) \Delta t + \frac{c_{MT} P_{MT}(t) \Delta t}{\eta_{MT}} \tag{5}$$

where C_{MT} is the generation cost of micro turbine in t; b_{MT} is the O&M cost coefficient; P_{MT} is the output power of MT in t; c_{MT} is the unit price of fuel; η_{MT} is the operating efficiency of MT, which is taken as 40%; Δt is the time interval, which is taken to be 1h; and T is the total scheduling period, which is taken as 24h.

Since the power response of MT is fast relative to hourly scheduling, the power output be expressed as:

$$P_{MT,min} \le P_{MT}(t) \le P_{MT,max} \tag{6}$$

where $P_{MT,min}$, $P_{MT,max}$ are the upper and lower limits of the power output of the MT, respectively.

4.2 Renewable Distributed Generation

O&M costs can be defined as:

$$C_j(t) = b_j P_j(t) \Delta t, \forall t \in T, \forall j \in \{PV, WT\} \tag{7}$$

where $C_j(t)$ is the O&M cost of WT and PV in t; b_j is the O&M cost coefficient; and $P_j(t)$ is the power output of RE in t.

4.3 Energy Storage Systems

Charge/discharge state constraints for ESS:

$$O_{ch}(t) + O_{dis}(t) \le 1 \tag{8}$$

where $O_{ch}(t)$ represent the charging and discharge state of the ESS in t.

Charge/discharge output power constraints of ESS:

$$0 \le P_{dis}(t) \le O_{dis}(t) P_{dis,max} \tag{9}$$

$$0 \le P_{ch}(t) \le O_{ch}(t) P_{ch,max} \tag{10}$$

$$P_{ESS}(t) = P_{dis}(t) - P_{ch}(t) \tag{11}$$

where $P_{ESS}(t)$ is the power output of the ESS at time t; $P_{dis}(t)$ represents discharge power of the ESS in t, $P_{ch}(t)$ represents charge power of the ESS in t.

State of charge (SOC) constraints for ESS:

$$SOC(t) = SOC(t-1) + \left(\frac{\eta_{ch} P_{ch}(t)}{P_r} - \frac{P_{dis}(t)}{\eta_{dis} P_r} \right) \Delta t \tag{12}$$

$$SOC(t=0) = SOC(t=T) \tag{13}$$

$$SOC_{\min} \le SOC(t) \le SOC_{\max} \tag{14}$$

where constraint (12) denotes the energy balance constraint for energy storage; η_{ch} and η_{dis} represent the charging and discharging efficiencies of the ESS, respectively; P_{r} denotes the rated capacity of the ESS; $SOC_{\min}$ and $SOC_{\max}$ represent the lower and upper limits of the ESS charging state, respectively. Constraint (13) ensures that the capacity of the ESS is equal at the beginning and last moment of scheduling, which facilitates the cyclic scheduling of the ESS, and constraint (14) represents the charge state constraint of the ESS.

Lead-acid and lithium batteries are frequently employed in ESS of MG, and the battery life depends on the depth of charge/discharge. When the depth of charge/discharge increases, the battery life decreases. The aging cost of ESS can be expressed as:

$$C_{\mathrm{ESS}}(t) = c_{\mathrm{ESS}}(P_{\mathrm{ch}}(t) + P_{\mathrm{dis}}(t))\Delta t, \forall t \in T \tag{15}$$

where $C_{\mathrm{ESS}}(t)$ is the aging cost of ESS at *t*; c_{ESS} is the aging cost coefficient.

4.4 Demand Response Load

Demand response (DR) is designed to encourage users to adjust loads that do not need to run to generate revenue.

$$P_{\mathrm{DR}}(t) = P_{\mathrm{up}}(t) - P_{\mathrm{down}}(t) \tag{16}$$

$$0 \le P_{\mathrm{up}}(t) \le O_{\mathrm{up}}(t) P_{\mathrm{up,max}} \tag{17}$$

$$0 \le P_{\mathrm{down}}(t) \le O_{\mathrm{down}}(t) P_{\mathrm{down,max}} \tag{18}$$

$$O_{\mathrm{up}}(t) + O_{\mathrm{down}}(t) \le 1 \tag{19}$$

$$\sum_{t=1}^{\mathrm{T}} P_{\mathrm{up}}(t) = \sum_{t=1}^{\mathrm{T}} P_{\mathrm{down}}(t) \tag{20}$$

where $P_{\mathrm{DR}}(t)$ represents the actual dispatch power of DR in *t*, meaning the amount of transferred electric loads; $P_{\mathrm{up}}(t)$ and $P_{\mathrm{down}}(t)$ denote the upward/downward adjustable power of DR in *t*, respectively. The power limits for upward and downward adjustment are given by constraints (17) - (18). Constraint (19) ensures that the load cannot be shifted downward and upward at the same time, $O_{\mathrm{up}}(t)$ represents the upward shifted state of the load at *t*, and $O_{\mathrm{down}}(t)$ represents the downward shifted state at *t*. Constraint (20) ensures that the total load demand remains constant during the scheduling period.

The scheduling cost of implementing DR can be expressed as:

$$C_{\mathrm{DR}}(t) = c_{\mathrm{DR}}(P_{\mathrm{up}}(t) + P_{\mathrm{down}}(t))\Delta t, \forall t \in T \tag{21}$$

where $C_{\mathrm{DR}}(t)$ is the implementation cost of DR in *t*; c_{DR} is the dispatch cost coefficient of DR.

4.5 Transmission Power with the Grid

To avoid the situation of MG purchasing and selling power to the grid at the same time, expressed as:

$$O_{\mathrm{buy}}(t) + O_{\mathrm{sell}}(t) \le 1 \tag{22}$$

$$0 \le P_{\mathrm{buy}}(t) \le O_{\mathrm{buy}}(t) P_{\mathrm{buy,max}} \tag{23}$$

$$0 \le P_{\mathrm{sell}}(t) \le O_{\mathrm{sell}}(t) P_{\mathrm{sell,max}} \tag{24}$$

where $O_{\mathrm{buy}}(t)$ denotes the state of MG purchasing power from the grid in *t*, and $O_{\mathrm{sell}}(t)$ denotes the state of MG selling power to the grid in *t*; $P_{\mathrm{buy}}(t)$ and $P_{\mathrm{sell}}(t)$ are the power purchased and sold to the grid in *t*, respectively. The interacting power limit with the grid is given by constraints (23) and (24).

The power balance constraint is:

$$\begin{aligned} &P_{\mathrm{buy}}(t) - P_{\mathrm{sell}}(t) + P_{\mathrm{ESS}}(t) + P_{\mathrm{PV}}(t) + \\ &P_{\mathrm{WT}}(t) + P_{\mathrm{MT}}(t) = P_{\mathrm{DR}}(t) + P_{\mathrm{RL}}(t) \end{aligned} \tag{25}$$

Where: $P_{\mathrm{RL}}(t)$ is the non-dispatchable load in the grid in time *t*.

The cost of buying and selling electricity interactively with the main grid is:

$$C_{\mathrm{G}}(t) = (c_{\mathrm{buy}}(t) P_{\mathrm{buy}}(t) - c_{\mathrm{sell}}(t) P_{\mathrm{sell}}(t))\Delta t \tag{26}$$

4.6 Pollutant Treatment Costs

The cost of MG to combat environmental pollution includes the environmental compensation cost of MT and the environmental compensation cost of fossil energy combustion in the distribution grid. The pollutants considered in this paper mainly include CO_2, SO_2, and NO_x, and their costs are described as follows:

$$\begin{aligned} C_{\mathrm{emi}}(t) = &\sum_{p} (q_{\mathrm{MT},p} P_{\mathrm{MT}}(t) + q_{\mathrm{G},p} P_{\mathrm{buy}}(t)) \\ &b_p \Delta t, \forall p \in \{\mathrm{CO_2}, \mathrm{SO_2}, \mathrm{NO}_x\} \end{aligned} \tag{27}$$

where $C_{\mathrm{emi}}(t)$ represents the pollutant treatment cost of the MG in t; $q_{\mathrm{MT},p}$ is the emission factor of pollutant type p generated by controllable distributed generation; b_p is the treatment cost of pollutant type p; and $q_{\mathrm{G},p}$ is the emission factor of pollutant type p generated by the distribution grid.

5 Two-Stage RO Model

5.1 Objective Function

In this paper, a two-stage RO model for MG is developed to minimize the daily operating cost and environmental costs, and the constraints to be satisfied include Eqs. (6), (8-14), (16-20), and (22-25). The objective function is shown in Eq.(28):

$$\min \sum_{t=1}^{T} \Big[C_{\mathrm{MT}}(t) + C_j(t) + C_{\mathrm{ESS}}(t) + C_{\mathrm{DR}}(t) + C_{\mathrm{G}}(t) + C_{\mathrm{emi}}(t) \Big] \tag{28}$$

The outcomes of the scheduling process derived from the deterministic optimization model are heavily influenced by the precision of the forecasted data. However, the inherent uncertainties present in the MG substantially affect the reliability of these scheduling outcomes. Consequently, it is imperative to incorporate these uncertainties into the development of the day-ahead dispatch strategy. The two-stage "min-max-min" robust optimization model constructed in this paper is shown in Eq. (29). The constraints include (6), (8-14), (16-20), (22-25).

$$\begin{cases} \min\limits_{x}[\max\limits_{u} \min\limits_{y} c^{\mathrm{T}} y] \\ s.t. Ax \geq d \\ By = e \\ Cx + Dy \geq f \\ Fy \geq h \\ Iy = u, u \in U_{\psi} \end{cases} \tag{29}$$

where the inner layer 'max' to search for the worst scenario with concentrated uncertainty, and 'min' is to minimize the operation cost of MG under the worst scenario. The outer layer 'min' is to optimize the scheduling by the corresponding dispatch power in the worst-case scenario. c denotes the column vector of coefficients corresponding to the objective function Eq. (28); B and I denote the equation constraints coefficient matrices; e denotes the constant column vector of equation constraints; A, C, D, and F denote the moments of unequal constraints coefficients of the day-ahead scenario; d, f, and h denote the constant column vectors of inequality constraints.

x and y denote the optimization variables with the following expressions:

$$\begin{cases} x = \begin{bmatrix} O_{\mathrm{buy}}(t), O_{\mathrm{sell}}(t), O_{\mathrm{ch}}(t), O_{\mathrm{dis}}(t), \\ O_{\mathrm{up}}(t), O_{\mathrm{down}}(t)) \end{bmatrix} \\ y = \begin{bmatrix} P_{\mathrm{MT}}(t), P_{\mathrm{DR}}(t), P_{\mathrm{ch}}(t), P_{\mathrm{dis}}(t), \\ P_{\mathrm{buy}}(t), P_{\mathrm{sell}}(t) \end{bmatrix} \end{cases} \tag{30}$$

5.2 Solution Algorithms

The two-stage RO problem cannot be solved directly. Therefore, the C&CG algorithm is adopted to solve robust optimization problems. The proposed two-stage RO model (29) needs to be firstly divided into a master problem (31) and a subproblem (32), and then solved by alternating iterations between the master problem and subproblem. The master problem is written in the following form:

$$\begin{cases} \mathrm{MP}: \min\limits_{x} \lambda \\ \lambda > c^{\mathrm{T}} y_k \\ s.t. Ax \geq d \\ By_k = e \\ Cx + Dy_k \geq f \\ Fy_k \geq h \\ Iy_k = u_k^*, u_k^* \in U^w \end{cases} \tag{31}$$

where U^w is the set of all searched worst cases. For each worst-case scenario u_s in U^w, generate the corresponding hourly phase variable y_k and its constraints. The mean value of the uncertainty range u_{mean} is chosen as the initial worst-case scenario u_0^* of uncertainty.

The subproblem can be expressed as follows:

$$\begin{cases} \mathrm{SP}: Sl(u, x^*) = \max\limits_{u} \min\limits_{y} c^{\mathrm{T}} y \\ s.t. Ax^* \geq d \\ By = e \\ Cx^* + Dy \geq f \\ Fy \geq h \\ Iy = u, u \in U \end{cases} \tag{32}$$

The inner layer of the subproblem contains only continuous variables, the max-min subproblem is transformed into a single-layer "max" problem utilizing the duality method. $\Omega(x, u)$ represents the feasible set of the optimization variable y given a set of (x, u), with the following expression:

$$\Omega(x, u) = \begin{Bmatrix} y| \\ Ax \geq d, \rightarrow \sigma \\ By = e, \rightarrow \omega \\ Cx + Dy \geq f, \rightarrow \tau \\ Fy \geq h, \rightarrow \theta \\ Iy = u, \rightarrow \rho \end{Bmatrix} \tag{33}$$

where $\sigma, \omega, \tau, \theta$ and ρ denote the dyadic variables corresponding to each constraint in the inner layer, respectively.

Given (x, u), the inner layer of the subproblem is linear, and the dual of the subproblem can be obtained from the correspondence in Eq. (33):

$$\begin{cases} \mathrm{SP}: \max\limits_{u \in U, \sigma, \omega, \tau, \theta, \rho} d^{\mathrm{T}}\sigma + (f - Cx)^{\mathrm{T}}\tau + \\ e^{\mathrm{T}}\omega + h^{\mathrm{T}}\theta + u^{\mathrm{T}}\rho \\ s.t. A^{\mathrm{T}}\sigma + B^{\mathrm{T}}\omega + D^{\mathrm{T}}\tau + F^{\mathrm{T}}\theta + \\ I^{\mathrm{T}}\rho \le c \\ \sigma \ge 0, \tau \ge 0, \theta \ge 0 \end{cases} \tag{34}$$

However, the existence of the bilinear term $u^{\mathrm{T}}\rho$ in the subproblem poses a challenge to the solution of the inner problem. Considering that the optimization result of u must be the vertex of each subinterval of the uncertainty set, the deviation variable is either 0 or 1. Eq. (34) is then transformed into a MILP problem by using the big M method: the form of the multiplication of binary and continuous variables will appear, and it will be linearized by introducing the auxiliary variables to linearize it.

$$\begin{cases} \mathrm{SP}: \max\limits_{u \in U, \sigma, \omega, \tau, \theta, \rho} d^{\mathrm{T}}\sigma + e^{\mathrm{T}}\omega + (f - Cx)^{\mathrm{T}}\tau + \\ h^{\mathrm{T}}\theta + \bar{u}\rho + \sum\limits_{j=1}^{N_{\mathrm{AS}}} (\varphi_{j+}\Delta u_{j+}) - \sum\limits_{j=1}^{N_{\mathrm{AS}}} (\varphi_{j-}\Delta u_{j-}) \\ s.t. A^{\mathrm{T}}\sigma + B^{\mathrm{T}}\omega + D^{\mathrm{T}}\tau + F^{\mathrm{T}}\theta + I^{\mathrm{T}}\rho \le c \\ 0 \le \varphi_{j+} \le M\pi^{j+} \\ 0 \le \varphi_{j-} \le M\pi^{j-} \\ \rho - M(1 - \pi^{j+}) \le \varphi_{j+} \le \rho \\ \rho - M(1 - \pi^{j-}) \le \varphi_{j-} \le \rho \\ \sum\limits_{t=1}^{T} \left(\varepsilon^{j+}_{\psi,t,s} - \varepsilon^{j-}_{\psi,t,s}\right) \le \Gamma_{\psi,j} \\ \sigma \ge 0, \tau \ge 0, \theta \ge 0 \end{cases} \tag{35}$$

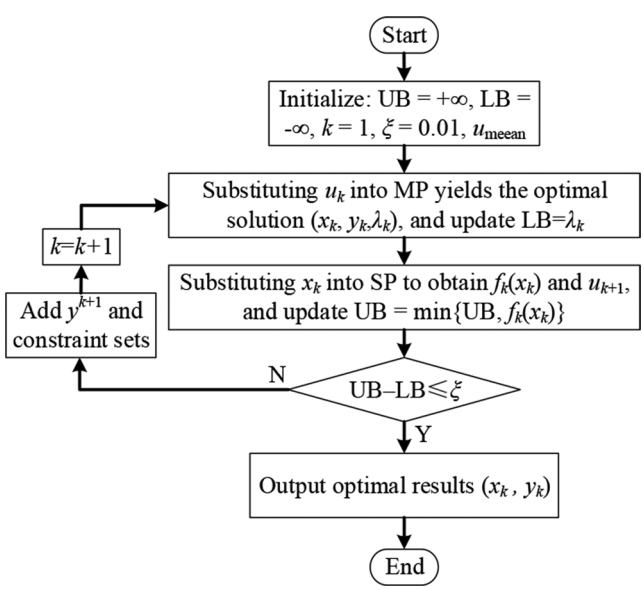


Fig. 7 Solving process of C&CG algorithm

where Δu_{j+}, Δu_{j-} represent the power prediction deviations; φ_{j+}, φ_{j-} are the introduced continuous auxiliary variables; π^{j+}, π^{j+} are the binary variables; and M is the upper bound of the dyadic variables.

After the above derivation and transformation, the proposed model has been transformed into a MILP problem, which can be solved by the C&CG algorithm iteratively for both the main problem and the subproblems until the difference between the LB and the UB is reduced to the convergence threshold ξ. Then the final optimized day-ahead scheduling decision is obtained. The C&CG algorithm solution flowchart is shown in Fig. 7.

1) Given a set of uncertain variables u_{mean}, as the initial worst-case scenario, $\mathrm{UB} = +\infty$, $\mathrm{LB} = -\infty$, the number of iterations $k = 1$, and the convergence gap is set to $\xi = 0.01$;

2) Substitute u_k into the master problem to solve the optimal results (x_k, y_k, λ_k) and update the objective function value of the master problem as the new $\mathrm{LB} = \lambda_k$;

3) Substitute the obtained x_k into the subproblem to get the optimal solution of the subproblem $f_k(x_k)$ and the corresponding worst case u_{k+1}, and update the value of the subproblem objective as the new upper bound $\mathrm{UB} = min\{\mathrm{UB}, f_k(x_k)\}$;

4) If $\mathrm{UB} - \mathrm{LB} \le 0.01$, stop the iteration and return the optimal results (x_k, y_k), otherwise, add the variable y^{k+l} and the following constraints:

$$\begin{cases} \lambda \ge c^{\mathrm{T}} y^{k+l} \\ Ax \ge d \\ By = e \\ Cx + Dy^{k+l} \ge f \\ Fy^{k+1} \ge h \\ Iy^{k+1} = u^{k+1} \end{cases} \tag{36}$$

Let $k = k + 1$ and return to step 2) until the algorithm converges. The overall solution flowchart of the proposed method is shown in Fig. 8.

6 Case Studies and Analysis

The developed RO scheduling model is modeled using MATLAB 2018a and YALMIP, and the CPLEX solver is invoked to solve the optimization problem.

6.1 Simulation Parameter Settings

To verify the validity and reasonableness of the proposed RO scheduling model for MG, simulations were performed based on the MG structure illustrated in Fig. 1. The simulation parameters are shown in Table 2. The emission factors

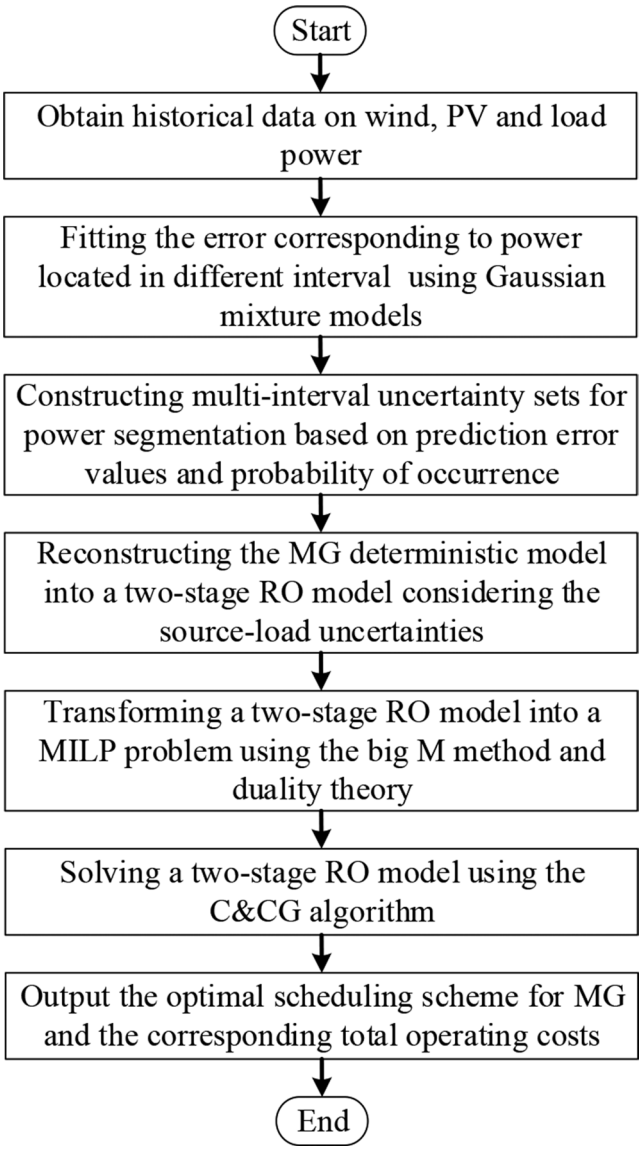


Fig. 8 The overall solution flowchart of the proposed method

Table 2 Parameter settings

Parameters	Value
$P_{\mathrm{MT,min}}$,$P_{\mathrm{MT,max}}$/kW	80, 800
b_{MT},c_{MT} /¥/kWh	0.12, 0.4
$\eta_{\mathrm{ch}}/\eta_{\mathrm{dis}}$	0.95
P_{r}/MWh	2
$SOC_{\mathrm{min}}/SOC_{\mathrm{max}}$	0.1,0.9
c_{ESS}/¥/kWh	0.38
P_{upmax}, P_{downmax}/kW	70
c_{DR}/¥/kWh	0.32
P_{buymin}, P_{buymax} /kW	1500

Table 3 Emission factors and treatment costs of pollutants

pollutants	CO_2	SO_2	NO_x
b_p /(¥/kg)	0.21	14.842	62.694
q_{MT}/(g/kWh)	184	0.001	0.619
q_{G}/(g/kWh)	889	1.8	1.6

Table 4 Day-ahead trading time-of-use tariffs

Time period	Electricity price (¥/kWh)
1:00-7:00,23:00-24:00	0.48
7:00-9:00,12:00-14:00,19:00-23:00	0.9
9:00-12:00,14:00-19:00	1.35

Table 5 Parameters of WT power prediction error probability distribution

Segmentation interval	Parameters
[0,0.3]	$w_1 = 0.291, \mu_1 = -0.008, \sigma_1 = 0.016$ $w_2 = 0.709, \mu_2 = -0.042, \sigma_2 = 0.033$
[0.3,0.6]	$w_1 = 0.191, \mu_1 = 0.010, \sigma_1 = 0.037$ $w_2 = 0.809, \mu_2 = -0.078, \sigma_2 = 0.095$
[0.6,1.0]	$w_1 = 0.800, \mu_1 = 0.006, \sigma_1 = 0.015$ $w_2 = 0.200, \mu_2 = -0.102, \sigma_2 = 0.089$

Table 6 Parameters of PV power prediction error probability distribution

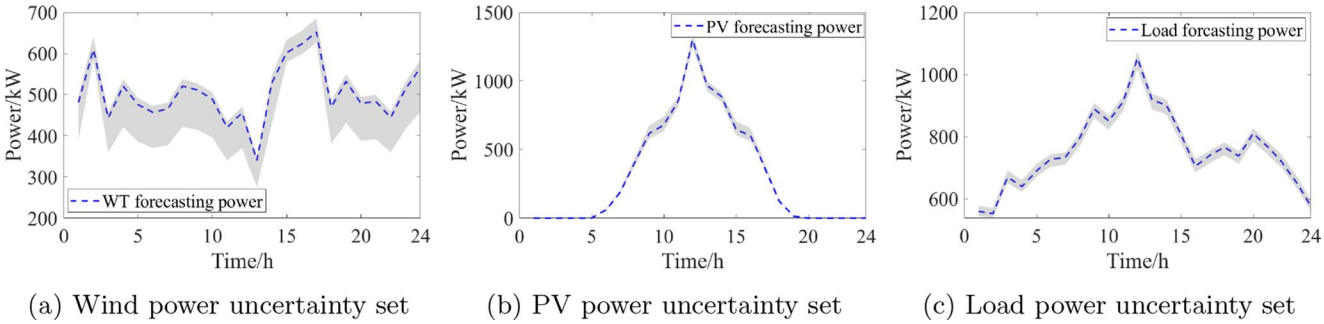
Segmentation interval	Parameters
[0,0.3]	$w_1 = 0.658, \mu_1 = 0.065, \sigma_1 = 0.014$ $w_2 = 0.342, \mu_2 = -0.008, \sigma_2 = 0.001$
[0.3,0.6]	$w_1 = 0.621, \mu_1 = 0.013, \sigma_1 = 0.026$ $w_2 = 0.379, \mu_2 = -0.009, \sigma_2 = 0.079$
[0.6,1.0]	$w_1 = 0.707, \mu_1 = -0.004, \sigma_1 = 0.013$ $w_2 = 0.293, \mu_2 = 0.008, \sigma_2 = 0.073$

Table 7 Parameters of load prediction error probability distribution

Segmentation interval	Parameters
[0,0.3]	$w_1 = 0.348, \mu_1 = -0.022, \sigma_1 = 0.011$ $w_2 = 0.652, \mu_2 = 0.002, \sigma_2 = 0.004$
[0.3,0.6]	$w_1 = 0.618, \mu_1 = 0.003, \sigma_1 = 0.002$ $w_2 = 0.382, \mu_2 = -0.011, \sigma_2 = 0.008$
[0.6,1.0]	$w_1 = 0.075, \mu_1 = -0.061, \sigma_1 = 0.005$ $w_2 = 0.925, \mu_2 = -0.005, \sigma_2 = 0.005$

and treatment costs of pollutants are shown in Table 3. Grid trading time-of-use tariffs are shown in Table 4.

According to the analysis in Sect. 3.1, the specific distribution parameters of the source-load segmented power prediction errors are shown in Tables 5, 6, and 7. According to the specific distribution parameters corresponding to the source-load segmented power prediction errors in different zones, the RE power uncertainty set is divided into three intervals with deviations of $\mu \pm \sigma$,$\mu \pm 2\sigma$, and $\mu \pm 3\sigma$, and the load uncertainty set is divided into two intervals with deviations of $\mu \pm 1.5\sigma$ and$\mu \pm 3\sigma$. The total interval Γ_ψ is divided according to the probability of occurrence of each interval, which can be set as: $(\Gamma_{\mathrm{pv},1}, \Gamma_{\mathrm{pv},2}, \Gamma_{\mathrm{pv},3}) = (4, 1, 1)$,



(a) Wind power uncertainty set (b) PV power uncertainty set (c) Load power uncertainty set

Fig. 9 Predicted source-load power curves

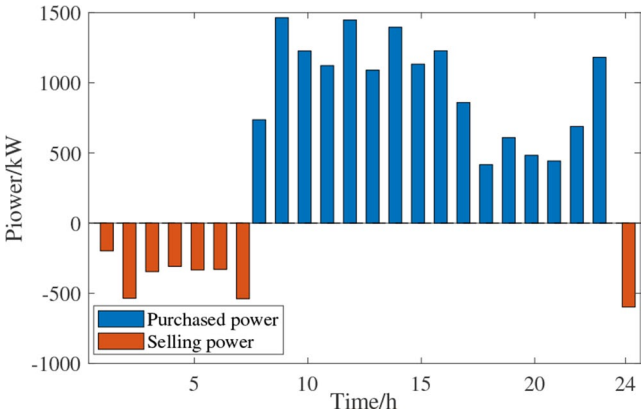


Fig. 10 Microgrid electricity sales program

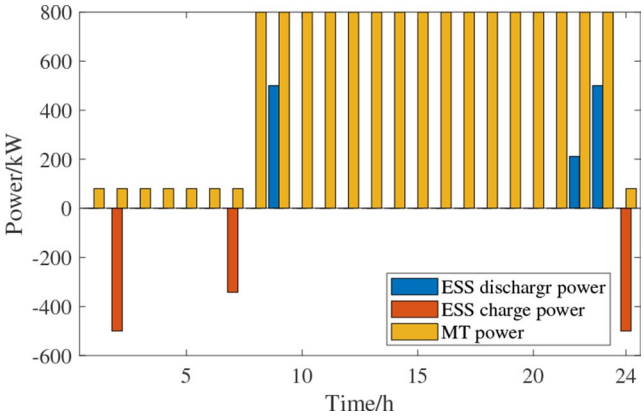


Fig. 11 Day-ahead scheduling plan for DG and ESS

$(\Gamma_{\mathrm{wt},1}, \Gamma_{\mathrm{wt},2}, \Gamma_{\mathrm{wt},3}) = (8, 3, 1)$, $(\Gamma_{\mathrm{L},1}, \Gamma_{\mathrm{L},2}) = (9, 3)$. The number and size of intervals depend on the amount of historical data obtained from actual projects. If more information about the dominant uncertainty is obtained, the uncertainty set can be divided into finer intervals, which is not only consistent with the actual situation; but also significantly reduces the conservativeness of the RO. Figure 9 demonstrates the predicted source-load power curves for the microgrid, with the shaded area indicating the range of uncertainty sets considered.

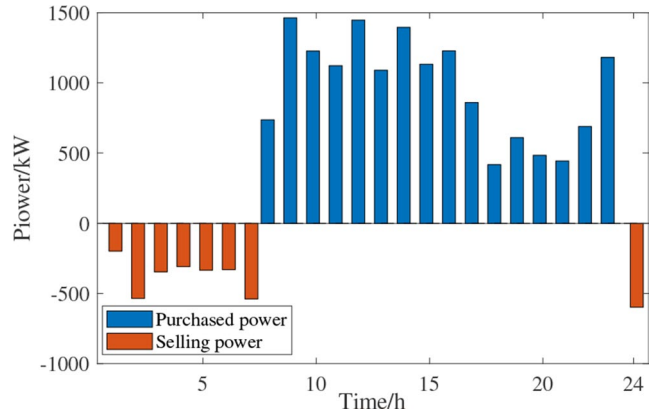


Fig. 12 DR actual demand electricity plan

6.2 Results and Analysis

MG was optimized and scheduled based on the proposed two-stage RO method. The transmitted power between the MG and main grid under the worst-case scenario is shown in Fig. 10. The charging and discharging power of the energy storage is shown in Fig. 11. The demand response results are shown in Fig. 12. In this case, the ESS discharges to the MG with the power represented by positive values, and the charging power is represented by negative values; the MG purchases power from the main grid with the power indicated by positive values, and the power sold is indicated by negative values.

As can be observed from Figs. 10 and 11, wind and PV energy generation is fully absorbed to maximize the use of clean energy. However, when the PV power generation is zero at night, renewable energy generation cannot meet the energy balance demand. To ensure the stable operation of the MG, the system adopts measures such as purchasing power from the grid, micro gas turbine power generation, and ESS power generation. At this time, the MT generation cost is higher than the grid electricity price, the MG mainly purchases electricity from the main grid, and the MT operates with minimum power output (24-next day 7h). During the rest of the time, the MT runs at maximum power

to increase the power sales to the distribution grid, which effectively increases the power sales revenue, enhances the energy interaction between the MG and the grid, and reduces the operating cost of the MG. The participation of DG effectively ensures that the MG sells power to the grid at maximum power during high tariff intervals, which can effectively reduce the operating cost. Meanwhile, the ESS is charged during low tariff periods and low load demand, and discharged during high tariff periods and high load demand, to realize the effect of peak shaving and valley filling in the grid. The participation of ESS effectively increases the absorption of renewable generation and reduces the operating cost of MG. The MG purchases power from the grid and charges the ESS in the low tariff interval. the MG sells power to the grid with maximum power in the high tariff interval. This reflects the role of the distribution network in supporting the economic and safe operation of the MG.

After the implementation of the demand response, the demand for electric loads changed. Figure 12 demonstrates the electric load demand before and after the implementation of the demand response. It can be seen that the electric load demand of the MG is shifted from the high tariff period to the low tariff period (1-7h) under the constraints of satisfying the total electric demand and electricity consumption, which contributes to the reduction of the cost of purchasing electricity from the main grid. Increasing the power output of the MT unit in the MG to reduce the cost of purchased power during high electricity price periods further optimizes the operational efficiency of the MG.

6.3 Comparison with Different Optimization Methods

Given the uncertainties of the RE and load, MG's two-stage RO model is developed. Four models are constructed in this section to characterize the accuracy of the proposed uncertainty set model directly for comparative experiments. Specifically as follows:

Model 1 (deterministic model): Uncertainties caused by source-load power is ignored in the scheduling and day-ahead forecast data is utilized; Model 2(SO): source-load uncertainties are described by a normal distribution; Model 3 (single-interval-based RO): uncertainties are represented by a traditional box-shaped uncertainties set; Model 4 (multi-interval-based RO): uncertainties are represented by a multi-interval uncertainties set; Model 5 (EUS-based RO): uncertainties are represented by an ellipsoidal uncertainty set(EUS); Model 6 (proposed): Uncertainties set are described by a data-driven segmented multi-interval uncertainties set.

Table 8 Cost comparison of different models

Model	Total MG cost/ ¥	Cost of carbon emissions / ¥	Solution time/s
Model 1	3645.8	565.8	4.38
Model 2	3814.5	580.6	225.28
Model 3	4259.5	622.8	4.42
Model 4	4038.7	608.4	4.73
Model 5	4450.9	649.6	8.75
Model 6	3912.3	595.3	105.83

The optimization results of the four models are shown in Table 8. From Table 8, it can be seen that Model 1 obtains the lowest scheduling cost and time used for computation because it does not consider the effect of uncertainties, computationally but due to the effect of uncertainties in the actual operation of the system, the scheduling results obtained are less robust. The total cost of running Model 2 is 4.6% higher than the deterministic model, but its solution time is significantly higher. This is because SO describes the uncertainties by generating a large number of randomized scenarios, which leads to an increase in the solution time, which will limit its application to complex scenarios. Model 3 has an 11.7% increase in the total cost of system scheduling compared to Model 2, and the solution time is comparable to that of Model 1, which indicates that RO is superior to SO in terms of computational efficiency, but its conservativeness leads to an additional cost increase. Model 4 reduces the total cost by ¥220.8 compared to Model 3, and the computation time is basically the same, which is due to the fact that the multi-interval uncertainty set further takes into account the prediction of uncertain parameters, which can better cope with the change of uncertainty and reduces the scheduling cost of the model. Model 6 compared with Model 4, the total cost decreased by ¥126.4 because the historical data distribution characteristics of the uncertainty parameters were considered and the uncertainty parameters were described more accurately to make them more realistic . However, the solution time is higher than other robust models, and the computational efficiency needs to be improved. The total cost of Model 5 is the highest among the six models, which is 4.5% higher than Model 3, and the computation time is shorter. This is due to the strict mathematical portrayal of correlation by the ellipsoidal constraints, which leads to poorer economy. Although the strategy of Model 6 has a higher total cost compared to the deterministic and SO models, Model 6 ensures that the system is risk-resistant, more robust, and better balances the economics and security of the MG.

6.4 Comparison of Microgrid Operating Costs with Different Uncertain Budget Parameters

To analyze the effect of uncertain budget parameters on the operating cost of MG, four uncertain budget parameters are set for comparison. The four uncertain budget parameters

are 0, 25%, 50% and 100%. The system dispatch costs under different uncertain budget parameters are shown in Table 9.

From Table 9, when the uncertainty budget parameters are all taken as 0, the robust optimization model is transformed into a deterministic optimization model, which results in the lowest total system scheduling cost. As the uncertainty budget parameters continue to increase, the total cost of scheduling the system also increases, and the scheduling scheme gradually tends to be conservative. The larger value of $\Gamma_{\psi,j}$ means that the uncertainty parameter takes more time periods to sub-boundary values in multiple intervals, and thus the uncertainty faced by the power balance of the microgrid is higher, leading to the more conservative scheduling scheme of the MG. Therefore, during the actual process of developing the dispatching scheme, appropriate uncertainty budget parameters can be set as needed to flexibly regulate the economy and robustness of the dispatching scheme.

Table 9 Effect of uncertain adjustment parameters on optimization results

$\Gamma_{\mathrm{pv},j}$	$\Gamma_{\mathrm{wt},j}$	$\Gamma_{\mathrm{L},j}$	Total MG cost/¥
(0,0,0)	(0,0,0)	(0,0)	3645.8
(2,1,0)	(4,1,1)	(4,2)	3842.5
(3,2,1)	(8,3,1)	(9,3)	3912.3
(6,4,2)	(16,6,2)	(18,6)	4154.3

7 Conclusion

In order to address the source-load uncertainties in microgrids, a two-stage RO scheduling model for MG based on data-driven uncertainty sets is developed considering the direct relationship between uncertainty parameter datasets and prediction errors. In the dispatch model, the MG operating cost and carbon emission cost are integrated. To reduce the conservativeness of the traditional RO, data-driven multi-interval uncertainty sets of power segments are introduced to describe the fluctuating intervals of uncertainties. The CC&G algorithm is used to solve the proposed model. The conclusions are drawn from the analysis of the case studies as follows:

(1) Considering demand response, carbon emission cost, and source-load uncertainty comprehensively, it can fully develop the scheduling potential of the source-load side, effectively smooth the RE and load fluctuations under the worst scenarios, and reduce the system operation cost.
(2) By DDSMIUS to describe the sources and load uncertainties, it can effectively reduce the conservativeness of the MG scheduling scheme and the total system dispatch cost. Compared with single-interval RO, multi-interval RO and ellipsoidal RO, the total cost of the proposed method is reduced by 8.15%, 3.13% and 12.10%, respectively, and the solution time is reduced by about 100s compared with SO.
(3) When the deterministic budget parameter setting is small, the scheduling cost of the system is low, but the robustness is weak; when the uncertainty parameter setting is large, the robustness of the system scheduling result is strong, but the scheduling cost is high. Therefore, the robustness and economy of the scheduling scheme can be flexibly adjusted by selecting appropriate uncertainty budget parameters according to the system demand during the actual formulation of the scheduling scheme.

In this paper, we focus on constructing an uncertainty set model for the RE and load uncertainties for traditional MG and solving a RO scheduling model for MG based on the proposed uncertainties set. Subsequent exploration and analysis will be carried out to explore and analyze a variety of complex uncertainties in integrated energy microgrids and multi-microgrids and carry out in-depth research on robust optimal scheduling of integrated energy microgrids and multi-microgrids.

Acknowledgements This work was supported in part by the National Natural Science Foundation of China (62072158, U2004163), the Key Research and Development Special Projects of Henan Province (231111221500), and Science and technology research project of Henan Province (232102210158, 242102210197).

Author Contributions Guodong Wang: Data curation, Methodology, Software, Formal analysis, Writing – original draft. Guoqiang Zheng: Conceptualization, Validation, Investigation, Writing – review \& editing, Funding acquisition. Zhonge Su: Methodology, Software, Formal analysis, Writing – review \& editing. Miaosen Hu: Supervision, Resources, Writing – review \& editing, Funding acquisition. Lingrui Kong: Supervision, Resources.

Data Availability Data will be made available on request.

Declarations

Competing Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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